9. (a) Find the derivative with respect to y of

$$\frac{1}{\left(1+2\ln y\right)^2}$$

(2)

(b) Hence find a general solution to the differential equation

$$3\text{cosec}(2x)\frac{\mathrm{d}y}{\mathrm{d}x} = y\left(1+2\ln y\right)^3 \qquad y>0 \qquad -\frac{\pi}{2} < x < \frac{\pi}{2}$$

(4)

(c) Show that the particular solution of this differential equation for which $y = 1$ at $x = \frac{\pi}{6}$ is given by

$$y = \mathrm{e}^{A\sec x - \frac{1}{2}}$$

where A is an irrational number to be found.

(5)